

## TANGIER IBN BATTUTA, Morocco

WMO: 601010

Lat: 35.727N

Lon: 5.917W

Elev: 62

StdP: 14.66

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 39.5       | 42.4 | 32.8                          | 27.4 | 48.1 | 35.7 | 30.8 | 48.3 | 31.4                  | 57.9 | 28.6 | 58.0 | 4.4                   | 90   | 0.496 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 16.3                   | 91.8            | 70.3 | 89.4 | 70.2 | 86.4 | 69.7 | 73.4                | 85.8 | 72.5 | 83.9 | 71.5 | 82.2 | 18.7                 | 60   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.7                            | 109.6 | 78.5 | 68.4 | 104.5 | 77.6 | 67.9 | 102.7 | 77.2 | 37.0          | 86.4 | 36.1 | 83.9 | 35.2 | 82.2 | 77.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 31.1              | 27.7 | 24.8 | 34.9                       | 99.3 | 3.4                | 4.0 | 32.5                                               | 102.2 | 30.5       | 104.5 | 28.6       | 106.7 | 26.1       | 109.6 |
|                   |      |      | 33.6                       | 75.3 | 3.4                | 1.2 | 31.2                                               | 76.1  | 29.2       | 76.8  | 27.3       | 77.5  | 24.8       | 78.4  |
|                   |      |      | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 65.0   | 54.6  | 55.3  | 58.4  | 61.2  | 65.9  | 72.0  | 75.2  | 76.7  | 73.2  | 69.0  | 61.1  | 57.0  | (6)  |
|                                                                        | DBStd       | 8.92   | 4.02  | 4.24  | 4.12  | 4.10  | 4.84  | 4.99  | 4.68  | 4.80  | 4.51  | 4.50  | 4.98  | 4.13  | (7)  |
|                                                                        | HDD50       | 20     | 8     | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 3     | (8)  |
|                                                                        | HDD65       | 1373   | 321   | 272   | 206   | 128   | 45    | 2     | 0     | 0     | 0     | 14    | 135   | 250   | (9)  |
|                                                                        | CDD50       | 5506   | 152   | 156   | 262   | 336   | 494   | 659   | 780   | 827   | 696   | 590   | 335   | 219   | (10) |
|                                                                        | CDD65       | 1382   | 0     | 0     | 2     | 14    | 74    | 211   | 315   | 362   | 246   | 139   | 19    | 0     | (11) |
|                                                                        | CDH74       | 9533   | 0     | 1     | 19    | 58    | 434   | 1456  | 2533  | 3047  | 1448  | 510   | 27    | 0     | (12) |
|                                                                        | CDH80       | 2991   | 0     | 0     | 3     | 4     | 86    | 467   | 904   | 1089  | 365   | 72    | 1     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 11.1   | 10.0  | 11.2  | 12.1  | 11.8  | 11.4  | 11.6  | 10.8  | 11.0  | 11.1  | 10.8  | 10.6  | 10.5  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 26.3   | 3.6   | 2.7   | 3.3   | 1.9   | 1.4   | 0.3   | 0.0   | 0.1   | 1.1   | 3.3   | 4.3   | 4.3   | (15) |
|                                                                        | PrecMax     | 62.0   | 18.6  | 11.0  | 9.1   | 5.9   | 4.5   | 2.6   | 0.4   | 0.5   | 2.5   | 8.6   | 11.7  | 18.5  | (16) |
|                                                                        | PrecMin     | 11.3   | 0.0   | 0.0   | 0.7   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.3   | 0.0   | (17) |
|                                                                        | PrecStd     | 9.8    | 3.6   | 2.4   | 1.9   | 1.3   | 1.2   | 0.6   | 0.1   | 0.1   | 0.7   | 2.2   | 3.2   | 4.2   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 67.9  | 69.9  | 76.0  | 78.9  | 86.3  | 93.6  | 95.4  | 96.8  | 90.7  | 84.6  | 77.3  | 70.2  | (19) |
|                                                                        |             | MCWB   | 59.0  | 59.7  | 62.0  | 62.9  | 65.2  | 67.6  | 70.2  | 70.5  | 69.6  | 67.0  | 63.7  | 59.2  | (20) |
|                                                                        | 2%          | DB     | 65.1  | 66.6  | 71.4  | 75.4  | 82.3  | 89.0  | 91.5  | 91.8  | 87.3  | 82.1  | 73.6  | 67.7  | (21) |
|                                                                        |             | MCWB   | 57.9  | 58.6  | 60.1  | 61.5  | 64.4  | 67.4  | 70.4  | 71.5  | 70.4  | 67.2  | 63.2  | 59.3  | (22) |
|                                                                        | 5%          | DB     | 63.9  | 64.6  | 68.3  | 72.1  | 78.8  | 85.5  | 88.4  | 89.5  | 84.3  | 79.1  | 71.4  | 65.8  | (23) |
|                                                                        |             | MCWB   | 57.5  | 57.8  | 58.8  | 60.8  | 63.7  | 67.3  | 70.2  | 71.4  | 69.5  | 66.6  | 62.8  | 59.0  | (24) |
|                                                                        | 10%         | DB     | 62.2  | 62.7  | 66.1  | 69.7  | 75.4  | 82.1  | 85.6  | 86.3  | 81.0  | 76.8  | 68.9  | 64.1  | (25) |
|                                                                        |             | MCWB   | 56.7  | 57.1  | 58.4  | 60.4  | 63.4  | 66.6  | 69.7  | 71.0  | 68.7  | 66.3  | 61.7  | 58.3  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 61.8  | 62.0  | 64.5  | 65.7  | 68.9  | 71.9  | 74.5  | 75.3  | 72.9  | 70.8  | 67.6  | 63.9  | (27) |
|                                                                        |             | MCDB   | 64.8  | 66.3  | 72.9  | 74.6  | 79.8  | 85.5  | 88.7  | 88.9  | 84.9  | 78.6  | 71.5  | 66.3  | (28) |
|                                                                        | 2%          | WB     | 59.9  | 60.3  | 61.9  | 63.9  | 66.7  | 70.2  | 72.9  | 73.8  | 71.9  | 69.3  | 65.3  | 61.9  | (29) |
|                                                                        |             | MCDB   | 63.4  | 64.6  | 67.8  | 71.8  | 76.9  | 82.3  | 85.6  | 86.2  | 82.6  | 77.1  | 70.8  | 65.0  | (30) |
|                                                                        | 5%          | WB     | 58.5  | 59.0  | 60.4  | 62.6  | 65.4  | 69.2  | 71.9  | 72.9  | 70.8  | 68.2  | 63.7  | 60.2  | (31) |
|                                                                        |             | MCDB   | 62.3  | 63.4  | 66.0  | 69.7  | 74.9  | 80.2  | 83.7  | 84.3  | 80.3  | 76.0  | 69.5  | 64.1  | (32) |
|                                                                        | 10%         | WB     | 57.2  | 57.6  | 59.3  | 61.1  | 64.3  | 68.1  | 70.8  | 72.0  | 69.8  | 67.1  | 62.3  | 58.9  | (33) |
|                                                                        |             | MCDB   | 61.3  | 62.1  | 64.9  | 67.8  | 73.2  | 78.1  | 81.5  | 82.6  | 78.4  | 74.5  | 67.7  | 63.1  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.4  | 14.6  | 15.0  | 15.2  | 16.3  | 16.4  | 16.8  | 16.3  | 14.7  | 14.2  | 13.6  | 13.5  | (35) |
|                                                                        |             | MCDBR  | 15.7  | 16.3  | 18.9  | 19.2  | 21.1  | 21.5  | 22.3  | 20.4  | 17.7  | 16.7  | 15.1  | 14.8  | (36) |
|                                                                        |             | MCWBR  | 10.6  | 11.0  | 11.3  | 10.9  | 10.1  | 8.5   | 8.7   | 7.8   | 7.3   | 7.6   | 8.6   | 9.5   | (37) |
|                                                                        | 5% WB       | MCDBR  | 13.7  | 14.6  | 16.8  | 17.4  | 19.1  | 18.5  | 19.4  | 18.0  | 15.4  | 14.7  | 13.2  | 12.5  | (38) |
|                                                                        |             | MCWBR  | 10.5  | 10.8  | 11.3  | 11.2  | 10.4  | 8.7   | 9.1   | 8.1   | 7.3   | 8.0   | 8.7   | 9.5   | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.337 | 0.362 | 0.406 | 0.402 | 0.418 | 0.441 | 0.448 | 0.458 | 0.430 | 0.398 | 0.364 | 0.343 | (40) |
|                                                                        | taud        |        | 2.506 | 2.429 | 2.285 | 2.327 | 2.309 | 2.253 | 2.222 | 2.202 | 2.314 | 2.404 | 2.473 | 2.504 | (41) |
|                                                                        | Ebn at Noon |        | 273   | 278   | 273   | 280   | 276   | 268   | 265   | 260   | 262   | 261   | 260   | 263   | (42) |
|                                                                        | Edh at Noon |        | 26    | 32    | 40    | 40    | 41    | 43    | 45    | 45    | 38    | 32    | 27    | 25    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 845   | 1110  | 1464  | 1838  | 2148  | 2410  | 2436  | 2154  | 1703  | 1243  | 903   | 754   | (44) |
|                                                                        | RadStd      |        | 79    | 126   | 141   | 104   | 140   | 100   | 51    | 71    | 75    | 107   | 73    | 71    | (45) |

## Historical Trends

|                        | DBAvg      | Heating    |            | Cooling    |            |              | Degree-Days |            |            |             |
|------------------------|------------|------------|------------|------------|------------|--------------|-------------|------------|------------|-------------|
|                        |            | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP        | HDD50       | HDD65      | CDD50      | CDD65       |
| Station Only           | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>-0.65 | (j)<br>N/A  | (k)<br>N/A | (l)<br>N/A | (m)<br>+196 |
| Regional (8 neighbors) | +0.34      | N/A        | N/A        | N/A        | N/A        | N/A          | N/A         | N/A        | N/A        | +103        |

Nomenclature: See separate page